EXPERIMENT NO: 06

Longest Substring Without Repeating Characters

1. Basic Information

Program Name: Experiment6.java

Package: recordprog

Language: Java

2. Objective / Aim

To develop a Java program that finds the longest substring in a given string without repeating characters and displays the substring and its length.

3. System Requirements

Hardware: Standard PC/Laptop; minimum 4 GB RAM; minimum 500 MB free storage.

Software: Windows/Linux/macOS; Java Development Kit (JDK); IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code; Command Prompt/Terminal.

4. Algorithm

1. Start the program.

2. Read a string from the user.

3. Initialize longest as an empty string.

4. For each starting position, initialize an empty current substring.

5. Add following characters while they are not already in current.

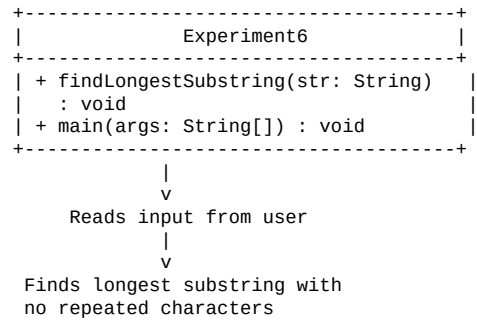
6. If a repeated character is found, stop the inner loop.

7. Update longest whenever current is longer than the longest found so far.

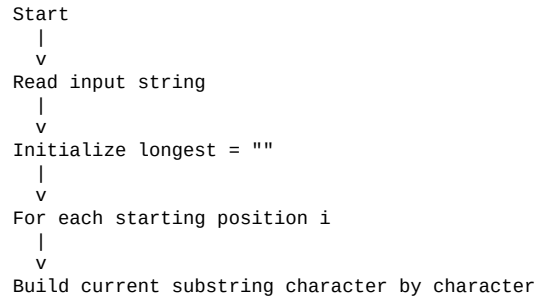
8. Display the longest substring and its length.

9. Close the scanner and stop.

5. UML / Program Structure



6. Flow of Execution



```
  |
  v
Is character already in current?
  | Yes                 | No
  v                     v
Stop inner loop       Append character
                        |
                        v
              Is current longer than longest?
                        |
                        v
                  Update if needed
                        |
                        v
               Continue searching
                        |
                        v
Print longest substring and length
  |
  v
Stop
```

7. Source Code

```
package recordprog;
import java.util.Scanner;

public class Experiment6 {

    // Method to find longest substring
    static void findLongestSubstring(String str) {
        String longest = "";

        for (int i = 0; i < str.length(); i++) {
            String current = "";

            for (int j = i; j < str.length(); j++) {
                char ch = str.charAt(j);

                // Check whether character is already present
                if (current.indexOf(ch) != -1) {
                    break;
                }

                current = current + ch;

                // Update longest substring
                if (current.length() > longest.length()) {
                    longest = current;
                }
            }
        }

        System.out.println("Longest substring: " + longest);
        System.out.println("Length: " + longest.length());
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        findLongestSubstring(str);

        sc.close();
    }
}
```

8. Compilation and Execution

```
javac -d . Experiment6.java
java recordprog.Experiment6
```

9. Sample Output

Example 1:

```
Enter a string: abcabcbb
Longest substring: abc
Length: 3
```

Example 2:

```
Enter a string: pwwkew
Longest substring: wke
Length: 3
```

10. Result

Thus, the Java program to find and display the longest substring without repeating characters, along with its length, was written and executed successfully.